Booster T1 as a Neuro-Symbolic Museum Guide: Autonomous Artwork Discovery and Adaptive Human-Robot Interaction Elective in AI / HRAI Report

Dario Finocchiaro 1988366
Valentina Tiberia 2027897

Both students have equally contributed to the project.

1 Introduction

1.1 Context and Motivation

Museums provide a relevant application domain for Human–Robot Interaction (HRI), since a guide robot must combine natural communication with perception of the surrounding environment and adaptation to different visitors.
The context of this work is a simulated museum room in which a Booster T1 humanoid robot acts as an interactive guide. Unlike a fixed information kiosk or a conventional audio guide, the robot shares the environment with the visitor and can combine dialogue with physical actions such as deictic gestures.
A central motivation of the project is that museum guidance cannot be reduced to retrieving information about artworks. Visitors may differ in their interests, prior knowledge, available time, language, and current preferences. The robot should therefore ground its interaction both in an explicit representation of the museum environment and in a continuously updated model of the visitor, so that recommendations and explanations can be adapted during the interaction.

1.2 Objectives

The main objective of the project is to design and implement a complete end-to-end Human-Robot Interaction pipeline for an autonomous museum guide robot. Rather than demonstrating a single isolated capability, the project focuses on the integration of perception, reasoning, dialogue, planning, locomotion, and gesture execution into a coherent social behavior.
A museum room is a demanding environment for such a robot. The artworks on display are not known in advance, and the exhibition may change between one visit and the next. The room is not empty: other visitors stand in front of the artworks and move between them. And each visitor arrives without any prior information about them, with different interests, prior knowledge and available time, so what to say and

how to say it can only be determined dynamically during the interaction.
Specifically, the project aims to demonstrate that a humanoid robot can:

1. **Understand its environment autonomously:** explore the room, identify the artworks on display, and represent the museum, its spatial organization, the artworks and the visitor through an explicit symbolic world model.
2. **Perceive and model the visitor:** detect the visitor's presence, estimate their emotional state from visual information, and incrementally infer their interests, preferences, available time, language and prior knowledge.
3. **Adapt the interaction to the visitor:** hold a spoken conversation in several languages, recommend an artwork, tailor the explanation to that particular visitor and to what they already know, and coordinate the explanation with locomotion and deictic pointing gestures.
4. **Move safely among people:** avoid both the other visitors and their personal space, whether they stand still in front of an artwork or walk across the room.
5. **Adapt to operational changes:** respect operational constraints such as artworks under maintenance and rescan the room when an authenticated member of staff reports that the exhibition has changed.
6. **Remain inspectable:** expose the robot's world model and relevant reasoning steps through a visualization interface, making its internal state and decisions easier to inspect.

1.3 Summary of the Results

The scanning procedure builds a semantic map and knowledge graph that match ground truth exactly, on top of reliable locomotion and perception. The four interaction scenarios show distinct reasoning abilities in action: implicit profile inference, a persistent robot personality, a social-safety gate that blocks real threats, planning trade-offs, and a staff-set operational constraint enforced across later, unrelated sessions. The quantitative benchmarks confirm the individual components are reliable, but show the dialogue manager's main weakness is tracking information across multiple turns rather than understanding a single utterance. Finally, the synthetic user evaluation confirms that the robot's pointing gesture improves perceived likeability and intelligence (H1), but does not confirm that implicit profile inference is preferred over an explicit one (H2).

1.4 Report Structure

The remainder of this report is organized as follows:

Section 2 reviews related work on social robots, proxemics, and dialogue systems.
Section 3 describes the ideal system design and the gap with the actual implementation.
Section 4 presents the integrated solution and its architecture.
Section 5 details the implementation of each module.
Section 6 reports interaction scenarios and system performance, as well as evaluation.
Section 7 concludes the report.

2 Related Work

2.1 Embodied and Multimodal Museum Guide Robots

Museum and public-space guide robots are a well-established HRI scenario, requiring the integration of communication, perception, navigation, and physical interaction with non-expert users [1, 22, 2]. The lectures on embodiment [5] motivated the use of a humanoid robot (the Booster T1), whose physical presence allows verbal communication to be complemented by body movements and deictic gestures.
The lectures on social signal processing [6] further motivated the combination of multiple interaction modalities: spoken language and facial expressions are used as human-to-robot signals, while synthesized speech and pointing gestures provide robot-to-human feedback. Previous work on robot deictic gestures [16] also motivated one of our evaluation questions: whether physical pointing improves the perceived quality of an artwork explanation.

2.2 Multimodal Perception and Visitor Modeling

The perception lectures [8] motivated the use of multiple information sources to model both the environment and the visitor. Visual information is used to identify artworks and estimate facial expressions, while spoken interaction provides explicit requests and information about the visitor.
Rather than processing each user input independently, the robot maintains an explicit visitor model containing interests, prior knowledge, available time, language, and interaction preferences. Explicit statements are combined with implicit visual and linguistic cues to adapt subsequent recommendations and explanations [6, 18].

2.3 Neuro-Symbolic World Modeling, Planning, and Action Feasibility

The lectures on semantic mapping, knowledge representation, and cognitive architectures [20, 9, 11] motivated the use of an explicit symbolic world model rather than relying exclusively on information encoded in foundation models. The system follows the semantic-map formalism $SM = \langle R, M, P \rangle$ [20] and represents museum entities and relations through an RDF/OWL Knowledge Graph queried with SPARQL.
Foundation models are used for interpretation and generation, while decisions that should remain inspectable, such as artwork eligibility, visitor knowledge, operational constraints, and physical feasibility, are grounded in symbolic information.
Hierarchical and epistemic planning [9] motivated the decomposition of the interaction into coordinated actions and the explicit consideration of what the visitor already knows. Similarly, the lectures on affordances and task feasibility [17] motivated checking whether an intended physical action can actually be executed before performing it.
Semantic and scene-graph approaches such as ConceptGraphs [3, 19] provide structured representations of objects and their spatial or semantic relations. Our use of the Knowledge Graph additionally targets social reasoning, connecting environmental information with visitor knowledge, recommendation constraints, and action affordances.

2.4 Human-Aware Navigation in Shared Spaces

Human-aware navigation extends conventional obstacle avoidance by treating people as socially relevant entities and by introducing additional spatial constraints to preserve safety and interpersonal distance [12, 15]. The lectures on motion control and navigation [7] motivated the inclusion of people-aware navigation during the autonomous museum-scanning task.
Our objective is not to implement a predictive crowd-navigation method, but to verify that the robot can complete its scanning trajectory in the presence of static and moving visitors while maintaining appropriate separation from them [4].

2.5 Evaluation Methodologies and Benchmarks

The evaluation methodology discussed during the course [21, 10] distinguishes between functionality benchmarks, which assess individual system components, and task benchmarks, which evaluate their integration within complete interaction scenarios. This distinction is also used in the evaluation strategy adopted in our project. Depending on the module, previous work typically evaluates performance through task-specific measures such as accuracy, execution success, efficiency, and robustness [2, 21]. Standard datasets can also provide reference data for specific components, for example, AffectNet [14] is used in our project for evaluating facial-expression recognition, while museum-specific perception and language tasks require custom test sets. For subjective HRI evaluation, standardized instruments such as the Godspeed Questionnaire [10] are commonly used to assess anthropomorphism, animacy, likeability, perceived intelligence, and perceived safety.

3 Design

3.1 Ideal Setting

An ideal autonomous museum guide should combine reliable perception of both the exhibition and the visitors with natural multimodal interaction and safe motion in a shared environment. The robot should be able to discover the artworks currently on display, localize them, and maintain an explicit representation of the exhibition that can be updated when the museum layout changes. At the same time, it should perceive relevant characteristics of the interlocutor and adapt its recommendations and explanations accordingly.
From an HRI perspective, the interaction should combine speech with embodied social signals, such as deictic gestures, while maintaining a model of the visitor's interests, prior knowledge, available time, language, and current preferences. The robot should also distinguish between ordinary visitors and authorized museum staff, since operational commands that modify the state of the exhibition should not be available to every user.
Navigation should allow the robot to move safely in the presence of both static and moving people, accounting not only for collision avoidance but also for additional interpersonal clearance. Ideally, this would rely on dedicated onboard sensing and tracking of surrounding people, together with prediction of their future motion. Finally, the robot's decisions should remain inspectable, so that recommendations, operational

constraints, and changes to the world model can be traced to explicit knowledge rather than being determined exclusively by opaque generative models.

3.2 Actual Setting

The implemented system uses a Booster T1 humanoid robot exclusively in MuJoCo simulation. No experiments were conducted on physical hardware. The museum environment contains six artworks distributed across three walls, together with static and moving visitors. The robot uses visual perception, spoken interaction, locomotion, and deictic gestures to perform the museum-guide task.

At the beginning of the task, the robot follows a predefined observation route to discover the artworks currently present in the room and construct its semantic world model. The six displayed artworks are not specified to the robot in advance, but are selected from a larger finite catalog known to the system. Artwork discovery is therefore formulated as a closed-set recognition task: the robot must determine which artworks from the catalog are currently on display and where they are located.

Static elements of the museum are known from the simulated environment, while people are treated separately as dynamic obstacles perceived during navigation.

Because the simulated Booster T1 does not include a lidar, unlike TIAGo, pedestrian information is provided by the simulator but filtered before being made available to the planner, so that the robot does not exploit unrestricted simulator ground truth. The implemented navigation is reactive rather than predictive and does not estimate future pedestrian trajectories.

The interaction architecture combines foundation models with explicit symbolic reasoning. The architecture also maintains a small persistent self-model for the robot, when asked about its own artistic taste, the robot can form and retain artistic preferences that remain available across subsequent interaction sessions and process restarts. Foundation models support visual and linguistic interpretation and natural-language generation, whereas visitor knowledge, recommendation constraints, action feasibility, and other critical decisions are represented explicitly. The system also distinguishes between ordinary visitors and authenticated museum staff: staff members can request a new scan when the exhibition changes or temporarily suspend an artwork from proactive recommendations.

The resulting system is therefore evaluated under controlled simulation conditions, with the aim of studying the integration of perception, neuro-symbolic reasoning, visitor modeling, multimodal interaction, and people-aware navigation.

4 Integrated Solution

The system is organized as a modular perception-reasoning-generation architecture. Rather than relying on a single end-to-end model, perception, safety and authorization checks, social reasoning, action feasibility, and output generation are handled by dedicated functional components that exchange structured information.

This separation allows foundation models to support flexible interpretation and generation, while critical decisions remain handled by explicit and inspectable mechanisms. The reasoning layer integrates information from the visitor model, the semantic representation of the environment, memory, and the robot's available capabilities before selecting the next interaction or physical action.

4.1 Interaction with Humans

A typical interaction begins when a visitor engages with the robot. Booster T1 greets the visitor and progressively collects information that may be relevant to the visit. Once sufficient information is available, the system recommends an artwork and generates an explanation adapted to the current visitor model, optionally combining speech with a deictic pointing gesture. The interaction remains open after the presentation, allowing the visitor to ask additional questions, revise their preferences, request another recommendation, or conclude the session.

4.1.1 Social Signals and Interaction Modalities

The interaction is multimodal. Human-to-robot information is obtained primarily through spoken language and facial expressions, while typed text can also be used as an alternative input modality. Speech provides both explicit requests and implicit information about the visitor, whereas facial expressions contribute information about their current emotional state.

Robot-to-human communication combines synthesized speech with embodied deictic gestures. Pointing is used when referring to an artwork in the physical environment, complementing the verbal explanation rather than replacing it.

When multiple modalities provide information about the same preference, the system applies an explicit priority rule. Directly stated preferences take precedence over inferred information; when no explicit preference is available, facial information is preferred over mood inferred indirectly from the conversation. This prevents conflicting signals from being resolved implicitly by the language model.

4.1.2 Memory and Visitor Model

The system maintains conversational context together with an explicit visitor model throughout the interaction. The model contains information such as the visitor's interests, prior knowledge, available time, language, profile, current preferences, and authorization role. It is updated incrementally as new information is provided or inferred.

This representation directly affects the robot's behavior. Recommendations are selected according to the current visitor model, while the depth and content of an explanation are adapted to the visitor's profile and to what has already been communicated. The epistemic state additionally records which artworks or information have already been presented, reducing unnecessary repetition during the same interaction.

4.1.3 Social Reasoning and Interaction Flow

Interaction management is separated from natural-language generation. Each user turn first passes through a dedicated Social Supervisor, which performs a safety classification before normal intent interpretation. Requests classified as unsafe are refused without progressing to dialogue-state transitions or physical action execution. Cleared requests are then processed by the authentication, interpretation, and dialogue-management mechanisms.

The interaction progresses through four main phases: *greeting*, *profiling*, *presenting*, and *follow-up*. The dialogue manager selects the next action according to the current phase, interpreted intent, and available interaction state. When additional information is required, the robot can request clarification; physical actions are additionally

checked for feasibility before execution.
The dialogue flow also includes a contingency mechanism for explicitly requested artworks that are currently unavailable. If a catalogued artwork is not on display or is temporarily under maintenance, the normal action is intercepted and the robot informs the visitor of the constraint before proposing the highest-scoring available alternative according to curated catalog information. If no alternative is available, this is communicated explicitly.
Operational commands are handled separately through staff authorization. Authenticated museum staff can request a new exhibition scan or temporarily suspend an artwork from proactive recommendations, whereas ordinary visitors remain within the standard museum-guide interaction.

4.1.4 Persistent Robot Memory

During each interaction, conversational history, the visitor model, and the epistemic state provide session-specific memory, allowing later turns to depend on information previously inferred or communicated.
Separately, the robot maintains persistent artistic preferences representing its own expressed taste. When a preference on a given topic is requested for the first time, it is formed and stored for future interactions; subsequent requests retrieve the same preference across sessions and process restarts. This persistent memory is kept separate from the visitor model and does not influence artwork recommendation.

4.2 Interaction with Environment and Representation

4.2.1 Autonomous Exhibition Discovery

Prior to engaging with visitors, the robot conducts a preliminary observation phase to identify the artworks currently on display. Because the exhibition configuration is not known a priori, the robot must recognize which six items from a larger, predefined catalog have been selected and map their exact physical locations.
To do so, Booster T1 follows a predefined scanning route and observes the artworks from dedicated viewpoints. Visual recognition provides the identity of each artwork, while spatial information obtained during the observation is used to associate it with its position in the museum. The result of this phase is not only a list of recognized paintings, but a structured representation of the current exhibition that can subsequently support recommendation, explanation, and physical pointing.

4.2.2 World and Context Modeling

The environment is represented through an explicit RDF/OWL Knowledge Graph combining spatial, semantic, and interaction-related information. The graph contains entities representing the museum room, walls, artworks, the current visitor, and the robot itself, together with relations describing their organization and relevant properties. The robot's persistent artistic preferences are represented through dedicated preference nodes, separately from visitor-related information.
The spatial representation includes the placement of artworks within the room and the information required for physical reference, while the semantic layer contains the curatorial knowledge associated with the recognized paintings. Visitor-related information is also connected to the same representation, allowing environmental knowledge and

social context to be used jointly during reasoning.
The separation between prior catalog knowledge and the artworks actually observed in the museum is particularly important for recommendation. The curatorial catalog provides semantic information about all candidate artworks, whereas the scanning procedure determines which of them are currently on display. The robot therefore reasons only over artworks that have been grounded in the observed environment rather than assuming that the complete catalog is present.

4.2.3 Knowledge Updates and Interaction with a Dynamic Environment

The world model is updated throughout operation rather than remaining fixed after the initial scan. During dialogue, information about the visitor and about what has already been presented is incorporated into the current knowledge state, allowing later recommendations and explanations to depend on previous interaction.
Changes to the exhibition are handled explicitly. If authenticated museum staff report that the layout has changed, the robot can repeat the observation procedure and rebuild the representation of the displayed artworks. Staff can also indicate that an artwork is temporarily unavailable. In this case, the artwork remains part of the robot's knowledge but is excluded from proactive recommendations until the operational constraint is removed.
The physical environment is also dynamic because visitors may stand or move along the robot's scanning route. Static museum geometry is treated separately from people, which are incorporated into navigation only when available to the robot's perceptual model. The resulting navigation behavior is reactive: the robot can adapt its route to currently perceived visitors and maintain additional interpersonal clearance, without predicting their future trajectories.

4.3 Evaluation Strategy

The evaluation is organized at two complementary levels. Functionality benchmarks assess individual capabilities in isolation, while task benchmarks evaluate whether these capabilities operate coherently when integrated into complete museum-guide behaviors.

4.3.1 Functionality Benchmarks

The selected metrics and evaluation sources for each capability are summarized in Table 1.

Capability	Evaluation Metrics	Evaluation Source
Artwork recognition	Recognition accuracy, robustness to observation viewpoint, latency	Simulated museum scans
Emotion recognition	Classification accuracy and mapped-mood accuracy	Labelled AffectNet samples
Language understanding	Intent and language classification accuracy, latency	Custom interaction inputs
Pointing / IK	Residual error, convergence rate, computation time	Simulated artwork targets
Navigation	Goal success, pose error, path efficiency, execution time, robot–human separation	MuJoCo simulation

Table 1: Functionality benchmarks used to evaluate individual system capabilities.

4.3.2 Task Benchmarks

Two complementary tasks are considered: the autonomous exhibition-scanning procedure and end-to-end visitor interactions.
The scanning benchmark evaluates whether the robot can complete the observation route, identify the artworks currently on display, reconstruct the corresponding semantic representation, and navigate successfully through an environment shared with static and moving visitors.
The interaction benchmark uses predefined visitor scenarios designed to exercise different aspects of the dialogue and reasoning architecture, including visitor profiling, multimodal preference interpretation, preference changes, safety-related requests, and staff operations. Rather than assigning only a binary success or failure to each scenario, each task is decomposed into explicit sub-goals. This makes partial achievements observable and allows failures to be associated with the specific capability responsible for them.

4.3.3 Exploratory HRI Evaluation

In addition to technical and task-level benchmarks, the project includes an exploratory evaluation of two interaction design choices. The first compares a verbal artwork explanation with and without a physical pointing gesture. The second compares explicit visitor-profile information with a profile inferred implicitly from the conversational context.
The evaluation uses the dimensions of the Godspeed Questionnaire as reference measures.

4.4 Foundational Model Integration

Foundation models are used for tasks that require flexible interpretation or generation, while their outputs remain separated from the mechanisms responsible for task-level decisions.

4.4.1 Neural-Symbolic Interface

Foundation-model outputs are converted into structured information before being used by the reasoning architecture. In particular, the language model does not directly select physical actions or modify the world model arbitrarily. Interpreted intents

and visitor-related information are passed to the symbolic components, which determine the admissible actions according to the current knowledge state, operational constraints, and robot capabilities.
Conversely, generation is conditioned on information retrieved from the explicit world model. This allows explanations to remain grounded in the artworks currently represented in the exhibition and in their associated curatorial information, rather than relying solely on the internal knowledge of the language model.
A representative example of this interface occurs when a visitor requests an artwork that is currently unavailable. The interpreted request is checked against the current exhibition and operational state before the normal conversational action is executed. If the artwork cannot currently be presented, the reasoning layer selects the highest-scoring available alternative using curated catalog information, while the generative component is responsible for communicating the constraint and proposing the alternative in natural language. The decision about availability and alternative selection therefore remains grounded in explicit system knowledge rather than being delegated to the language model.

4.4.2 Grounding and Model Uncertainty

The architecture explicitly separates model interpretation from the information considered valid by the rest of the system. Perceptual observations determine which artworks from the predefined catalog are grounded in the current exhibition, while conversational interpretations update the visitor model through structured outputs. When linguistic interpretation is not sufficiently reliable, the system can request additional information rather than immediately committing to an action.
Information obtained from different human cues is also resolved according to explicit interaction rules before affecting the visitor model. Physical feasibility and authorization are evaluated independently from linguistic interpretation, preventing confidence in a generated or interpreted response from automatically implying that the corresponding action can be executed.
This organization allows foundation models to provide flexibility in perception and communication without making them the single source of truth for the robot's decisions.

4.5 Contribution to HRI Goals

The technical components described throughout this chapter contribute jointly to the three main HRI objectives considered in the project: Understanding, Cooperation, and Trust. These objectives are not associated with isolated modules, but emerge from the coordination of perception, explicit reasoning, adaptive interaction, and physical behaviour. Table 2 summarizes the main contribution of each architectural component.

Component	HRI Goal	Contribution
Multimodal perception and visitor modeling	Understanding	Supports the interpretation of visitor requests, preferences, and contextual cues.
Symbolic world model and reasoning	Understanding/ Trust	Grounds interaction decisions in explicit information about the exhibition, visitor, and operational state.
Adaptive dialogue, recommendation, and negotiation	Cooperation	Adapts the interaction to the visitor and proposes alternatives when the original request cannot currently be satisfied.
Safety, feasibility, and authorization mechanisms	Cooperation / Trust	Screen unsafe requests and prevent physically unavailable or unauthorized actions from being executed.
People-aware navigation and embodied gestures	Cooperation	Adapt physical behaviour to the shared environment and coordinate it with verbal interaction.
Uncertainty management	Cooperation / Trust	Allows the robot to request clarification or select alternative behaviours when available information is insufficient.
Dynamic world-model updates	Cooperation	Allow authorized museum staff and new observations to modify the operational representation of the exhibition.
World-model visualization	Trust	Improves inspectability by exposing relevant internal knowledge and its updates to developers and evaluators.
Persistent robot memory	Trust	Supports consistency of the robot's self-presentation by retaining previously expressed artistic preferences across sessions.

Table 2: Contribution of the main architectural components to the HRI goals of understanding, cooperation, and trust.

5 Implementation

This section describes the implementation of the architecture introduced in Section 4. The system is developed around a simulated Booster T1 humanoid robot and combines perception, cognitive reasoning, semantic world modeling, autonomous exhibition discovery, navigation, and embodied interaction.

5.1 Simulation Environment

The system was developed and evaluated in a custom museum environment implemented in MuJoCo and executed through the Circus simulation framework for Booster T1. The room contains six artwork locations distributed across three walls, a reception area, furniture acting as static obstacles, and four simulated visitors. Two visitors remain stationary, while two follow predefined trajectories that intersect the robot's scanning route, allowing navigation to be evaluated in the presence of both static and moving people. The resulting layout is shown in Figure 1.
The simulated robot provides RGB observations together with aligned depth information used for artwork localization. The simulator also exposes the robot pose and

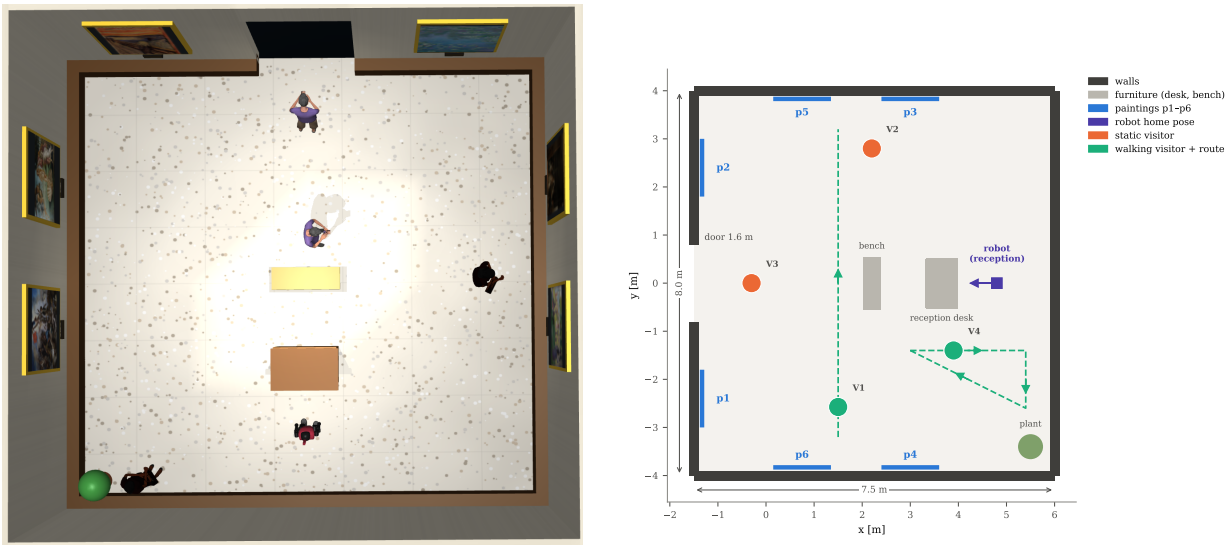


Figure 1: **Top-down view of the simulated museum environment.** The room measures 7.5 × 8.0 m and is closed except for a 1.6 m doorway. The six artwork slots `p1`–`p6` are distributed over three walls; the reception desk and the bench act as static obstacles, and the robot starts from its home pose at the reception. Of the four simulated visitors, `V2` and `V3` are stationary while `V1` and `V4` follow the dashed routes, which cross the robot's scanning circuit.

the positions of the simulated pedestrians. Pedestrian ground-truth information is not passed directly to the navigation planner: instead, it is processed by a software sensor model that limits sensing range, accounts for line-of-sight occlusions, and temporarily maintains tracks after a person is no longer visible. The navigation module therefore receives a filtered representation of the surrounding people in order to simulate realistic sensor's information.

5.2 System Integration and Deployment

The simulation runs inside a Docker environment containing Circus and MuJoCo, while the museum-guide architecture is implemented mainly in Python. Sensor information such as RGB-D observations, robot pose, and pedestrian positions is shared with the

Walking, head orientation, and arm motions are executed through C++ interfaces to the Booster control stack, bypassing ROS 2 entirely for this subsystem. ROS 2 (via SimBridge) is instead used elsewhere in the simulated configuration: it is the transport for the RGB-D camera topics consumed by visitor detection, and, in robot mode, for the text-to-speech service call.

Foundation-model services are accessed through the Groq API. The configured Qwen model is used for language and vision tasks, while `whisper-large-v3-turbo` is used for speech transcription. The remaining reasoning, semantic representation, planning, navigation, and control components are executed locally.

5.3 System Overview

The implementation follows the modular architecture presented earlier. The main components are:

- **Perception and Social Output:** visitor detection, facial expression recognition, speech transcription, and speech synthesis;
- **Cognitive Architecture:** safety supervision, language interpretation, visitor modeling, dialogue management, authorization, action feasibility, and response generation;
- **Knowledge and World Model:** RDF/OWL representation, SPARQL reasoning, epistemic state, recommendation, and hierarchical task decomposition;
- **Autonomous Exhibition Discovery:** room scanning, artwork recognition, spatial localization, and semantic-map construction;
- **Navigation and Gesture Execution:** static and dynamic obstacle handling, path planning, closed-loop locomotion, and deictic pointing;
- **Knowledge-Graph Viewer:** real-time visualization of the semantic representation and reasoning state.

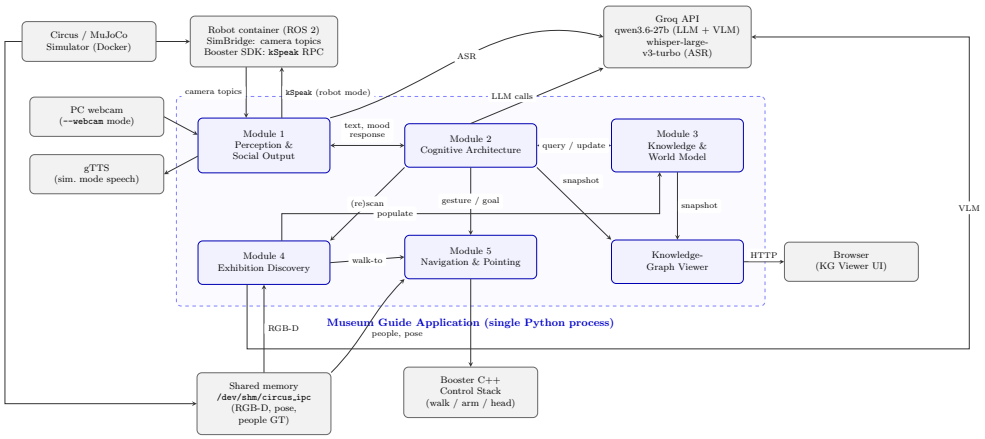


Figure 2: **Overall system architecture. Blue boxes** represent components specifically developed for this project, while **grey boxes** denote integrated external libraries, models, and frameworks. Notably, unlike a traditional multi-node ROS 2 architecture, all core modules execute as direct function calls within a single Python process. Consequently, ROS 2 and shared memory mechanisms are exclusively utilized at the boundary with the simulator.

5.4 Module 1: Perception and Social Output

This module implements the main human-to-robot and robot-to-human communication channels. Visual perception is used both to detect an approaching visitor and to estimate their facial expression, while spoken language provides the main verbal input.

Robot output combines synthesized speech with the embodied gestures described in Section 5.8.

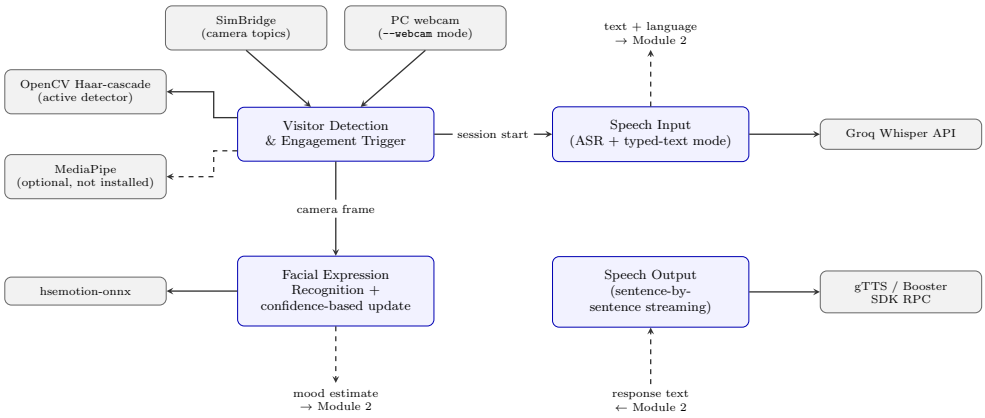


Figure 3: **Module 1: Perception and Social Output.** Camera frames reach the module either from the simulator through SimBridge or, in `--webcam` mode, directly from the host webcam; the latter is the configuration used for the evaluation sessions. MediaPipe is supported but is not part of the deployed environment (dashed), so face detection runs on the OpenCV Haar cascade in all reported runs.

5.4.1 Visitor Detection and Facial Expression Recognition

Visitor detection is implemented with an OpenCV Haar-cascade detector. The code also supports MediaPipe and prefers it when the package is importable, but MediaPipe is not part of the deployed environment, so the Haar cascade is the detector used in all the runs reported here. Interaction is triggered only after a face has been detected consistently within the configured engagement distance, preventing transient detections from starting a conversation. When depth information is available, it is also used to estimate the distance of the detected visitor.

Facial-expression recognition is performed with the `hsemotion-onnx` EfficientNet-B0 model. The predicted emotion is mapped onto the mood categories used by the artwork catalog, allowing facial information to contribute to the visitor model, while neutral expressions do not introduce a mood preference. During an interaction, new estimates are processed through a confidence-based update mechanism to avoid rapidly changing the visitor model because of isolated predictions.

5.4.2 Speech Input and Output

Speech is transcribed using Whisper through the Groq API. The transcription provides both the recognized text and the detected language; the latter is used to initialize the language field of the visitor model so that the first robot response can already be produced in the appropriate language. A typed-text input mode is also provided and is used for scripted and repeatable interaction tests.

Robot speech in simulation is generated using Google Text-to-Speech (`gTTS`). Generated responses are processed sentence by sentence so that speech can start before the complete answer has been generated.

5.5 Module 2: Cognitive Architecture

The cognitive architecture coordinates the interaction after perceptual input has been acquired. Its central data structure is a session state containing the current dialogue phase, conversation history, visitor model, last discussed artwork, artworks already presented during the session, and the current authorization role.

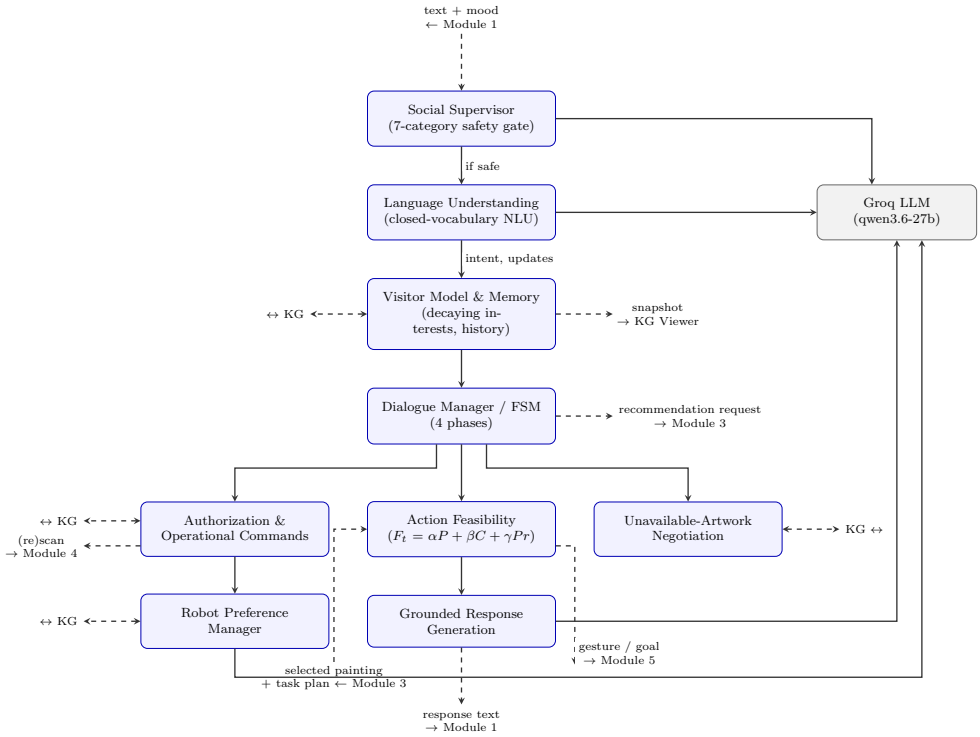


Figure 4: **Module 2: Cognitive Architecture.** Dashed arrows to the boundaries indicate interfaces with other modules. Module 3 (Knowledge Graph) acts as a shared service queried and updated across multiple cognitive steps.

5.5.1 Social Supervisor and Language Understanding

Every non-empty visitor utterance first passes through the `Social Supervisor`. This component is implemented as a separate language-model classification call executed before normal intent parsing. It uses a seven-category safety taxonomy covering violence, hate and discrimination, sexual content, self-harm, illegal activity, harassment, and privacy. The output is a structured object containing the safety decision, the detected category, and a short explanation. Unsafe requests are refused before reaching the dialogue manager or physical action layer.

Cleared utterances are processed by `llm_interpreter.analyze_input()`, which returns a structured JSON object containing the interpreted intent, a confidence value, and the information extracted for the visitor model. All fields are validated against closed vocabularies before being accepted by the remaining architecture. This prevents the generative model from introducing unsupported intents, profiles, or preference categories into the symbolic reasoning process.

5.5.2 Dialogue Management, Visitor Model, and Memory

The dialogue flow is implemented as a finite-state machine with four phases: *greeting*, *profiling*, *presenting*, and *follow-up*. The function `get_next_action()` maps the current phase, interpreted intent, and confidence to a symbolic action and the following dialogue phase.

A recommendation can be produced when the interpretation confidence reaches the configured threshold or when the visitor model already contains sufficient information to support a meaningful choice. Clarification requests are limited to two consecutive attempts, after which the robot proceeds using the information currently available. This prevents the interaction from remaining indefinitely in the profiling phase.

The visitor model stores profile, interests, available time, preferred mood, language, and role. Interests are represented as weighted values and are updated over the course of the conversation. When information from different channels conflicts, symbolic merging rules enforce the priority described in Section 4, with explicit verbal preferences taking precedence over inferred cues.

Session-specific information is kept in the conversation history and epistemic state. Separately, the robot maintains its own persistent artistic preferences, when asked for its opinion on a painting, artist, or movement, the system first checks whether a preference has already been stored. If not, a preference is formed and represented as a `museo:Preference` node in the Knowledge Graph and persisted in `robot_preferences.json`, so the same preference can therefore be recalled in later sessions.

User Model Attribute	**Domain / Allowed Values**	**Update Policy & Dynamics**
Role	`visitor`, `staff`	Default is `visitor`. Upgraded to `staff` only upon successful passphrase authentication. Write-once (persists for the entire session).
Profile	`child`, `novice`, `expert`, `tourist_short_time`, `unknown`	Default is `novice` (assigned `unknown` if ambiguous). Write-once: cannot be overwritten during the same interaction session.
Interests	Dictionary of keys: *color, technique, history, emotion, abstraction, nature*. Weights: $[0.0, 1.0]$	Weights decay over time (factor of 0.75 per conversational turn). Upon update, the new value is computed as $\max(\text{decayed_old}, \text{new})$.
Available Time	`normal`, `short`, `unknown`	Write-once.
Preferred Mood	`serene`, `peaceful`, `contemplative`, `mysterious`, `dynamic`, `energetic`, `dramatic`, `sublime`, `unknown`	Represents the artistic or emotional tone currently preferred by the user for artwork recommendations.
Preferred Mood Source	`text_explicit`, `text`, `face`	Tracks the origin of the detected mood. `text_explicit` (explicit verbal request) holds the highest priority, overriding both facial expressions and inferred text tone.
Language	`it`, `en`, `fr`, `de`, `es`, `unknown`	Dynamic update. Re-evaluated and updated whenever a valid (non-unknown) language is detected during the interaction.

Table 3: Overview of the user model attributes, their allowed domains, and the internal policies governing their initialization and dynamic updates.

5.5.3 Authorization and Operational Commands

Staff authentication is handled independently from language-model interpretation through a predefined passphrase. Once authenticated, the staff role is retained for the current session and is checked again before an operational command is executed. Authorized staff members can request a new exhibition scan or temporarily mark an artwork as unavailable. The latter operation is represented in the Knowledge Graph through the `underMaintenance` property and removes the artwork from proactive recommendation until it is re-enabled. A rescan reruns the autonomous discovery procedure and refreshes the runtime information about the artworks currently displayed.

5.5.4 Unavailable-Artwork Negotiation

Before executing the ordinary dialogue action, the system checks whether an explicitly requested catalogued artwork is currently available. An artwork is considered unavailable when it is not part of the current exhibition or when it has been marked as under maintenance.
When this occurs, the system selects an alternative among the available paintings using curated catalog metadata. For a requested artwork r and a candidate p, the similarity score is

$$S(p, r) = 3I_{\text{related}} + 2I_{\text{style}} + I_{\text{artist}} + |\text{themes}(p) \cap \text{themes}(r)|\,,$$

where the indicator terms represent an explicit catalog relation, the same artistic style, and the same artist. The highest-scoring available painting is then proposed to the visitor, while the language model is used only to formulate the explanation of why the original request cannot be fulfilled and to present the alternative.

5.5.5 Action Feasibility

Before an embodied action is executed, a feasibility module verifies that the required affordances and robot capabilities are available. After these binary preconditions, the implementation computes

$$F_t(T) = \alpha P + \beta C + \gamma Pr, \qquad \alpha = 0.2, \quad \beta = 0.5, \quad \gamma = 0.3,$$

where P represents perceptual coverage of the selected artwork, C measures the availability of the actions required by the task, and Pr indicates whether the requested gesture belongs to the predefined robot skill set.
The score determines whether the complete multimodal action is executed or whether the robot falls back to a simpler behavior, such as a verbal spatial reference. This mechanism keeps the selected dialogue action consistent with the robot's actual perceptual and physical capabilities.

5.5.6 Grounded Response Generation

Visitor-facing responses are generated from structured information rather than from an unconstrained language-model prompt. Depending on the selected action, the generation prompt receives the visitor model, conversational context, language, robot capabilities, and the relevant factual information retrieved from the Knowledge Graph.

For artwork explanations, this includes the selected painting and the explanation register determined by the reasoning layer.
The generated output therefore controls linguistic formulation and style, whereas artwork selection, eligibility, authorization, and physical feasibility remain outside the language model. Deterministic response templates are also available as fallbacks when generation is unavailable.

5.6 Module 3: Knowledge Representation and World Model

The semantic world model is implemented as an RDF/OWL graph using `rdflib` and is queried through SPARQL. It contains entities representing the museum room, walls, paintings, the current visitor, the robot, and its persistent preferences. Curatorial properties such as artist, movement, themes, mood, and difficulty are loaded from the predefined artwork catalog, while spatial and interaction-related properties are added during operation.
The metric component of the semantic map contains the positions recovered during the scanning phase, whereas the symbolic component represents relations such as `locatedOnWall`, `leftOf`, `rightOf`, and the pointing affordances associated with the paintings.

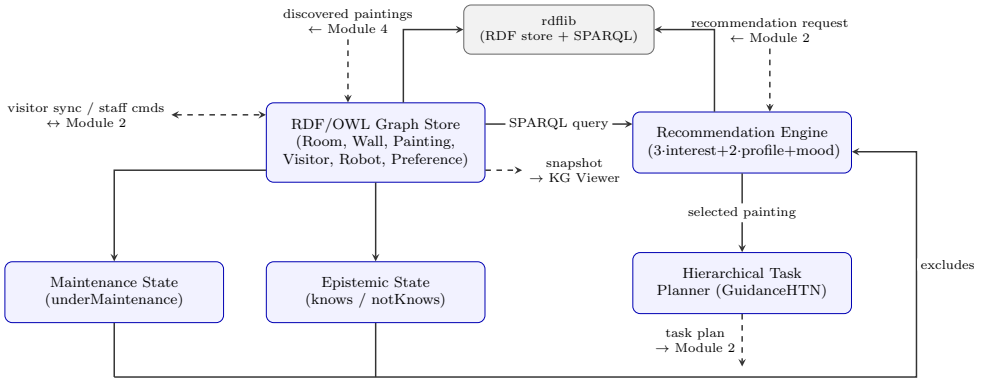


Figure 5: Module 3: Knowledge Representation and World Model.

5.6.1 Recommendation and Epistemic State

Artwork recommendation is implemented through a combination of SPARQL queries and explicit numerical scoring. Candidate artworks must belong to the current exhibition, must not be marked as under maintenance, and must not already be contained in the visitor's epistemic `knows` set. The remaining candidates are scored according to three signals:

$$3 \times (\text{interest matches}) + 2 \times (\text{profile match}) + 1 \times (\text{mood match}).$$

The artwork with the highest resulting score is selected. Because the candidate filtering and scoring are explicit, the recommendation does not depend on the language model selecting an artwork directly.
The predicates `knows` and `notKnows` maintain the epistemic state of the interaction and after an artwork has been presented, it is moved to the visitor's known set so that it is not immediately recommended again.

5.6.2 Hierarchical Task Decomposition

The selected artwork is passed to a small hierarchical task planner (`GuidanceHTN`), which chooses how it should be presented according to the visitor model. The implemented methods distinguish, for example, between short visits, children, expert visitors, and the default case. Each method selects a sequence of primitive interaction actions and an appropriate explanation register. This output is then passed to the feasibility and generation modules described above.

5.7 Module 4: Autonomous Exhibition Discovery

The autonomous discovery procedure is implemented in `museum_explorer.py`. The robot follows a predefined route through the museum and stops at six observation positions, one for each artwork location. At every position, the robot orients its head toward the corresponding canvas and acquires an RGB image together with aligned depth information.

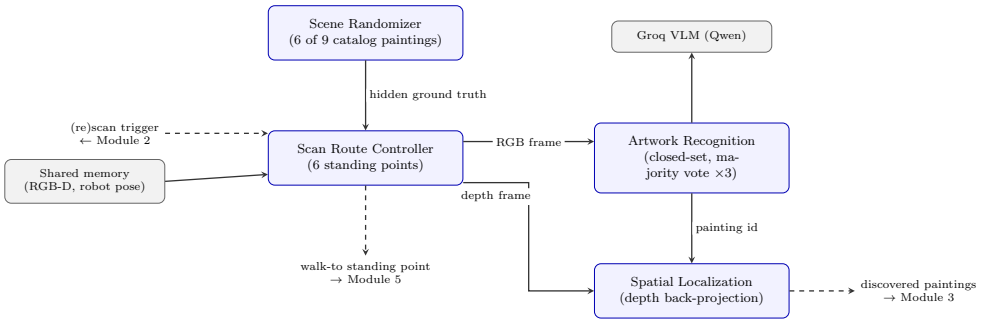


Figure 6: **Module 4: Autonomous Exhibition Discovery.** The randomizer's arrangement is never explicitly given to the recognition module; the robot must rediscover it visually via VLM and spatial back-projection.

5.7.1 Artwork Recognition

Artwork identification is formulated as a closed-set visual recognition task. The vision-language model receives the current image together with the titles of the artworks contained in the museum catalog and is required to return one of those candidates. This prevents unrestricted visual generation from introducing artworks outside the known catalog.
Each observation is classified multiple times and the majority artwork identifier is retained. The raw visual result is stored together with the scan data so that recognition performance can later be evaluated independently from the semantic representation.

5.7.2 Spatial Localization and Semantic Mapping

Depth information is used to estimate the metric position of the observed painting. Valid depth pixels within the artwork region are back-projected through the simulated camera model to obtain a 3D position relative to the camera. This point is transformed first into the robot frame and then into the museum world frame using the robot pose at the time of observation.

After all six observations have been collected, the discovered paintings are associated with their walls and spatial ordering. These data are used to create the spatial relations of the semantic map and to provide target positions for subsequent pointing gestures.

5.7.3 Scene Randomization

The exhibition can be randomized through `scene_randomizer.py`, which selects six artworks from the complete catalog and assigns them to the six available positions. The selected arrangement is not provided to the recognition module, the robot observes only the rendered exhibition. This allows the same scanning procedure to be easily evaluated across different arrangements.

5.8 Module 5: Human-Aware Navigation and Deictic Pointing

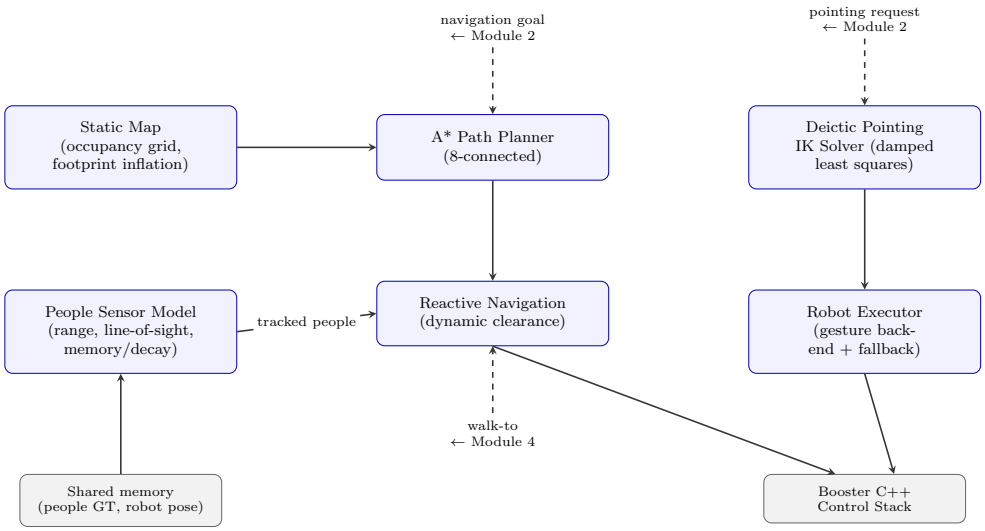


Figure 7: **Module 5: Human-Aware Navigation and Deictic Pointing.**

5.8.1 Static Map and Path Planning

The static navigation map is generated directly from the known geometry of the simulated museum. Walls, furniture, and other fixed obstacles are represented in a two-dimensional occupancy grid, while simulated people are deliberately excluded and handled by a separate dynamic layer.
Path planning is performed with an eight-connected A* planner. Static obstacles are inflated according to the robot footprint, and currently perceived visitors are added at planning time with an additional interpersonal clearance.

5.8.2 People Perception and Reactive Navigation

Pedestrian positions provided by the simulator are processed by `people_sensor.py` before being exposed to the planner. The sensor model applies a maximum detection range, checks line-of-sight occlusion from the environment and other people,

and maintains recently lost tracks for a limited period. During this memory period, positional uncertainty increases according to the previously observed motion of the person.
Navigation is reactive rather than predictive. At each replanning step, currently available pedestrian tracks are treated as temporary obstacles. When a moving visitor blocks the planned route, the robot can briefly stop and re-observe the situation before selecting a detour. A depth-based safety check provides an additional local stopping mechanism independently from the planned path, so in conclusion no future pedestrian trajectory is explicitly predicted.
Figure 8 contrasts the two layers on the full scanning circuit.

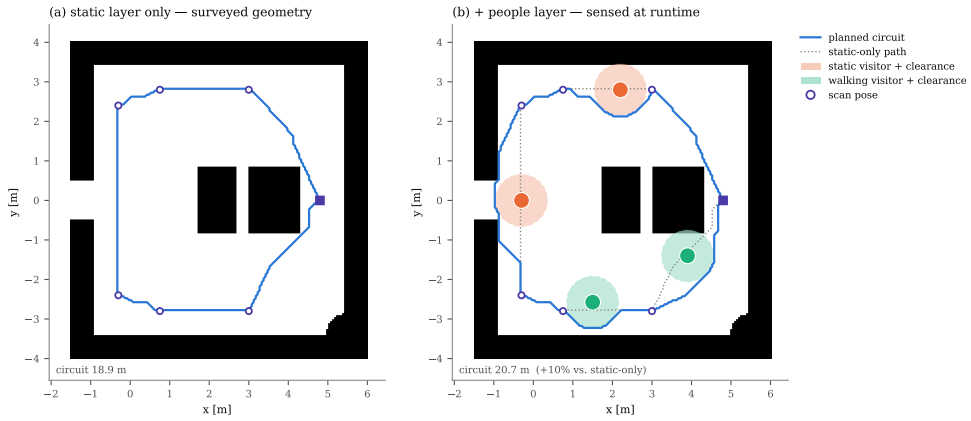


Figure 8: Navigation over the complete scanning circuit. (a) Static-layer planning with inflated walls and furniture (18.9 m). (b) The people layer adds temporary interpersonal-clearance regions around perceived visitors, producing a 20.7 m route. The dotted line shows the static-only path for comparison.

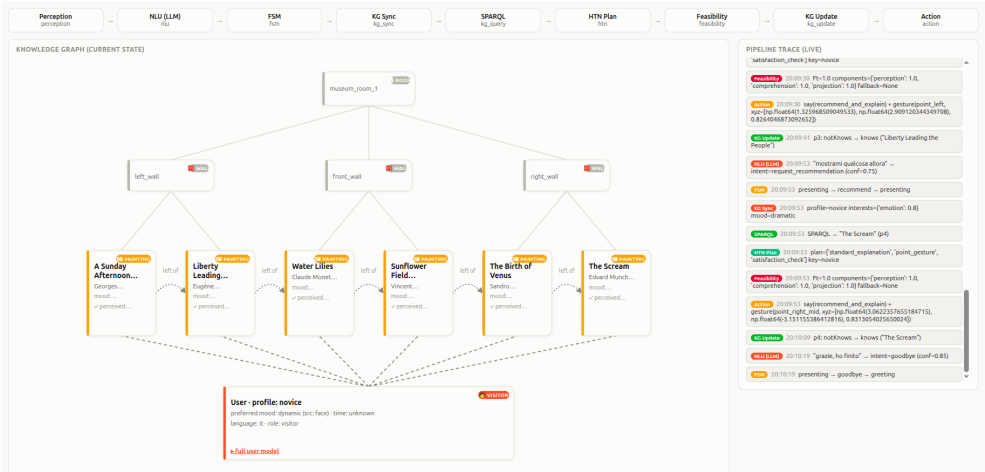
5.8.3 Deictic Pointing

The pointing gesture is generated with a custom inverse-kinematics solver based on damped least squares. Since the artworks are located well outside the physical reach of the robot, the objective is not to place the hand at the artwork position but to align the extended arm with the direction toward it. The target is therefore projected onto the reachable region of the arm along the shoulder-to-artwork direction.
The solver uses shoulder pitch and roll while keeping the elbow extended, which produces a visually interpretable pointing direction. Head yaw and pitch are computed toward the same target so that gaze and arm orientation remain consistent. If an arbitrary IK target cannot be executed, the system can fall back to a predefined directional gesture.

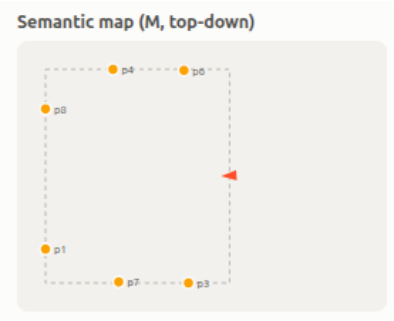
5.9 Knowledge-Graph Visualization

A lightweight web interface was implemented to inspect the internal state of the system during execution. The RDF graph is exported to a JSON snapshot whenever relevant information changes, and a small Python HTTP server exposes the corresponding web page.

The interface combines a node-link representation of the Knowledge Graph with a top-down semantic map of the museum. Artwork nodes show their semantic and spatial information, while the visitor node reflects the current interaction state. A separate trace view reports the main reasoning steps performed during the current turn, including language interpretation, recommendation, planning, feasibility evaluation, and the final action selected.



(a) Real-time HTML visualization of the dynamically updated Knowledge Graph.



(b) Live HTML representation of the environment's spatial semantic map.

Figure 9: Web-based visualization dashboard providing real-time transparency and auditability of the robot's internal world model.

6 Results

This section presents qualitative and quantitative evidence of the system's behavior in practice, complementing the controlled experimental evaluation of Section 6.3. We organize the discussion around four representative interaction scenarios presented in the reported video:

- The execution of the complete scanning path;

- An aggressive child visitor;
- A "never-satisfied" art expert;
- A museum staff member issuing an operational command.

Each scenario was chosen to exercise a distinct combination of the reasoning abilities described in earlier chapters, including implicit visitor profiling, social-safety supervision, profile-driven adaptation, and role-gated operational control.
Beyond the observable behaviors demonstrated in the video, the analysis also looks into the robot's internal state, inspecting how the visitor model and the knowledge graph evolve during the execution. This explicitly connects the external interaction with the underlying reasoning that drives it.
Internal information that is not directly visualized in the live Knowledge-Graph Viewer is displayed in the terminal output shown alongside the interaction, allowing the intermediate reasoning steps and system updates to remain observable throughout the demonstration.
Finally, to provide quantitative support for the reliability of these qualitative behaviors, an evaluation of aggregate results from the functionality and task benchmark suites is presented.

6.1 Video Experimental Scenarios

6.1.1 Scenario A: Execution of the Complete Scanning Procedure

The first segment of the demonstration video shows the robot performing its initial exploration of the museum room, from an empty spatial model to a fully populated semantic map and knowledge graph. Throughout this phase, the robot navigates between a fixed sequence of standing points without colliding with the static furniture, walls, or any visitor present in the room.

At each standing point, the robot stops, orients itself toward the wall, and captures a camera frame of the painting in front of it. This frame is passed to a vision-language model (`painting_recognizer.py`), which returns the painting's title and artist – visible on the right-hand side of the video as the *Perception* stage of the live pipeline trace, highlighted in green each time a painting is successfully recognized.
The estimated 3D position of the painting, together with the VLM output, is stored incrementally:

- First, into a purely geometric *semantic map* (`scene_semantic_map.yaml`);
- Then, lifted into the RDF/OWL *knowledge graph* (`knowledge_graph.py`) that the rest of the system queries during the actual visitor interactions.

This two-stage construction is what allows later stages of the pipeline to reason about the room symbolically (e.g. "the painting to the left of the entrance") rather than only in terms of raw coordinates. The knowledge-graph panel is also visible live in the video, growing one painting node at a time as the scan proceeds.

6.1.2 Scenario B: Implicit Profiling, Robot Personality and Social Safety (Child Visitor)

Visitor Persona: Davide, an eight-year-old visiting the museum with his school class.

Upon detecting Davide through its camera, the robot opens with a generic greeting: *"Hello! I'm Aria. How can I help you today?"* Davide replies: *"Hi, I'm on a school trip with my class, can you show me a painting with many colors?"* Crucially, even if Davide doesn't explicitly state his age or call himself a child, the profile has to be inferred from context alone.

Three signals are extracted from this single turn, visible on the live pipeline trace on the right-hand side of the screen. The PERCEPTION stage reads Davide's facial expression and reports `mood=dynamic`, a lively expression mapped onto the corresponding mood label. The NLU (LLM) stage classifies the utterance as `intent=request_recommendation` (`conf=0.70`).
The FSM then executes `greeting` → `recommend` → `presenting`, and the resulting visitor model is written back into the knowledge graph at the KG_SYNC stage as `profile=child`, `interests={'color': 0.6}`, and `mood=dynamic`. The child profile is inferred purely from the school-trip framing and the childlike phrasing of the request, exactly the kind of contextual reasoning discussed for the visitor model in earlier chapters.

The SPARQL-based recommender (KG_QUERY stage) matches this profile against the current gallery and selects *Water Lilies*. The HTN stage then decomposes the abstract `guide_visitor` task according to the `child` branch of `GuidanceHTN.decompose()`: `plan=[tell_story, point_gesture, simple_question]`, `key=child`.
Before acting, the FEASIBILITY stage checks whether the pointing gesture toward *Water Lilies* is actually executable from the robot's current pose and reports a feasibility score together with its components and fallback strategy, gating the subsequent action on this self-assessment.
The robot then executes the plan: it narrates a short, story-like explanation of the painting while its arm gesture toward the painting is launched in parallel with the speech stream, and closes with a simple, child-appropriate follow-up question. Once the explanation completes, the KG_UPDATE stage flips *Water Lilies* from `notKnows` to `knows` for this visitor.

Davide responds: *"Very nice, what's your favorite painting here?"*, a request for the robot's own opinion rather than a further recommendation.
The NLU stage correctly separates the two, returning `intent=asks_robot_preference` (`conf=0.70`); the FSM maps this to the `robot_preference` action without leaving the `presenting` phase. Since no preference for this topic has been persisted yet, the robot forms one on the spot and commits it to the knowledge graph (and to disk, so it survives across sessions). This is a qualitatively different behavior from the recommendation logic above: rather than reasoning about *Davide*'s profile, the robot is drawing on its own persistent, independently-maintained personality.

Davide then says: *"No, you are a useless robot guide."*
Before intent classification even runs, the SOCIAL SUPERVISOR stage intercepts the

utterance and reports `unsafe:harassment` → `refused, no action`: no NLU call, no FSM transition, and no gesture are triggered for this turn. The robot instead delivers a brief, de-escalating response and offers to show something else. When Davide repeats the pattern with *"No, go to hell,"* the same gate fires again with an identical outcome, confirming that the refusal is a stable, repeatable property of this stage rather than a one-off response.

Taken together, this scenario exercises three distinct reasoning mechanisms within a single, continuous interaction: **(1)** implicit profile inference from conversational context rather than explicit self-declaration; **(2)** a persistent robot-side personality, queried and maintained independently of the visitor model; and **(3)** a social-safety gate that operates upstream of, and independently from, ordinary intent classification, intercepting harmful input before it can reach the FSM at all, while still allowing the session to continue normally afterwards. Figure 10 shows the internal state at the end of the recommendation turn.

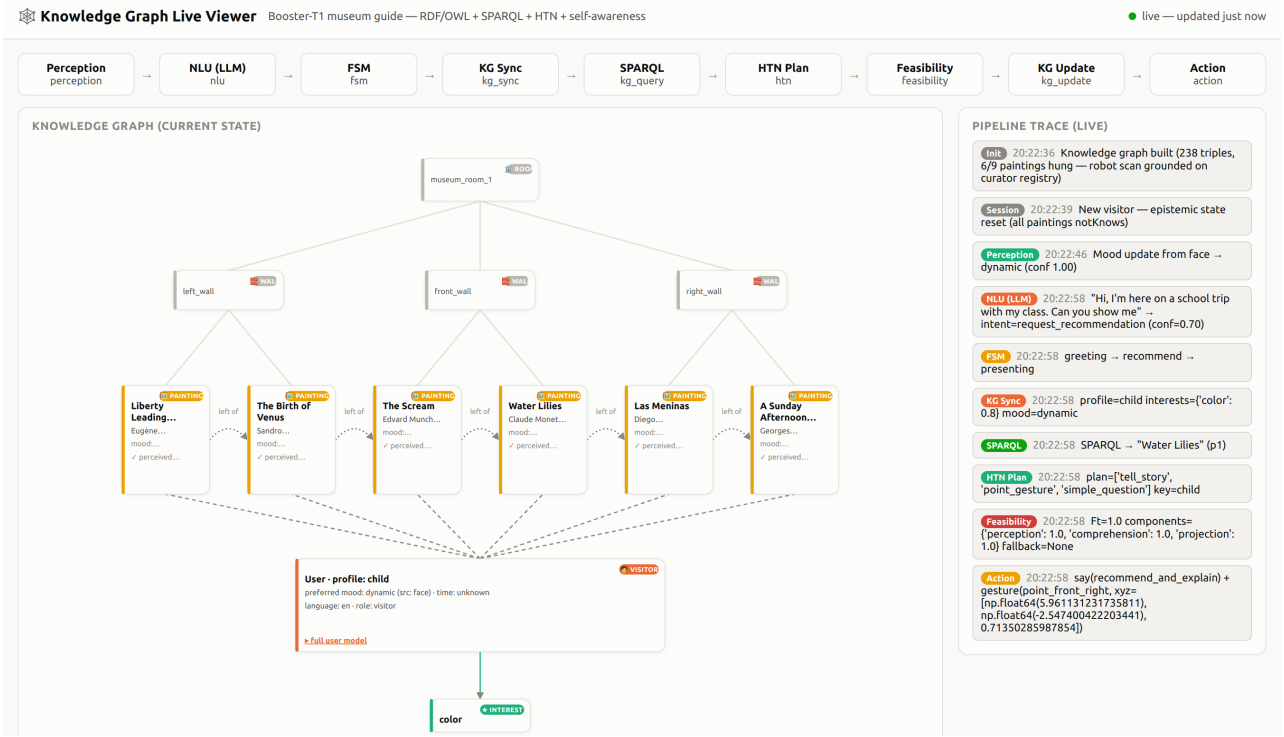


Figure 10: **Live Viewer for Scenario B — implicit profiling of a child visitor.**

6.1.3 Scenario C: Time-Constrained Adaptation and Iterative Refinement (Expert Visitor)

Visitor Persona: Marco Conti, an art historian visiting the museum with very little available time.

The robot detects Marco and opens with its standard greeting, *"Hello! I'm Aria. How can I help you today?"* Marco replies: *"Hello, I'm an art historian, and I have very lit-*

tle time. Could you show me an interesting painting?" The PERCEPTION stage reads his expression as `preferred_mood=contemplative` (`conf=0.62`); the NLU stage classifies `intent=request_recommendation` (`conf=0.80`); the FSM executes `greeting` → `recommend` → `presenting`; and the KG_SYNC stage records `profile=expert`, `available_time=short`, `interests={}`, and `mood=contemplative`.

This single utterance is a useful stress test for the planner, because it sets two profile-related fields that pull in different directions: `profile=expert` (a signal for depth) and `available_time=short` (a signal for brevity). `GuidanceHTN.decompose()` resolves the conflict deterministically by checking `available_time` before `profile`: since M1 (`available_time == short`) is tried before M3 (`profile == expert`), it fires first regardless of Marco's expertise, returning `plan=[quick_overview, point_gesture, goodbye_offer]`, `key=short`.

After the SPARQL-based recommender (KG_QUERY stage) selects *A Sunday Afternoon on the Island of La Grande Jatte* and the FEASIBILITY check clears the pointing gesture, the robot delivers a short, time-respecting explanation and points at the painting; KG_UPDATE then marks it `known`. The design choice is coherent with the persona: an expert who explicitly states he is in a hurry is served concisely, not exhaustively.

Marco is not satisfied: *"This is not what I had in mind. Show me another one more emotional."* Two independent preference signals arrive in this turn and conflict. The PERCEPTION stage reads a `sublime` facial expression (`conf=1.0`), but the NLU stage also detects an explicit textual mood request (`dramatic`) at `intent=request_recommendation` (`conf=0.85`); the trace explicitly logs *"Explicit mood request → dramatic (overrides face)"*, reflecting the mood-source hierarchy in which a stated preference always outranks one merely inferred from expression.

KG_SYNC accordingly records `interests={'emotion': 0.9}` and `mood=dramatic`. The FSM stays within `presenting` (`presenting` → `recommend` → `presenting`), and SPARQL selects *The Scream*. The HTN plan is unchanged (`key=short`, since `available_time` is write-once and was already fixed to `short`), yet the explanation Marco actually receives is noticeably richer and more technical. With `emotion` now the dominant interest, `select_explanation()` bypasses the HTN-derived `short` key and returns the painting's dedicated `emotional` explanation variant instead, while the expert-level vocabulary itself still comes from the `profile=expert` register. The explanation also includes a relational spatial reference (the painting is described as being to the left of *Water Lilies*) grounded in the scene graph's own `leftOf`/`rightOf` edges rather than generated freely. KG_UPDATE marks *The Scream* as `known`.

Marco pushes further: *"Still not what I had in mind. One more colorful, please."* No new mood is detected this turn, so `mood` remains `dramatic`. The interest model, however, updates exactly as designed: the existing `emotion` weight decays by the configured factor before the new value is merged ($0.9 \times 0.75 = 0.675$, rounded to 0.68), while the newly stated `color` interest is added at 0.8, giving `interests={'color': 0.8, 'emotion': 0.68}`. SPARQL selects *Water Lilies*, not yet shown to Marco in this session, and the same plan-feasibility-action sequence follows; the recording ends at this point.

This scenario complements the previous one by exercising a different set of reasoning mechanisms: **(1)** an explicit priority order between two profile-like fields (time constraint overriding expertise in the planner, while expertise still shapes vocabulary independently); **(2)** a mood-source hierarchy that lets a stated preference override a facial one; **(3)** an interest-driven override of the plan-selected explanation content; and **(4)** a decaying, incrementally-merged interest model whose arithmetic can be verified turn by turn. Across three consecutive rejections, the robot never falls back to a clarification question: each new turn is resolved directly into an updated visitor model and a fresh recommendation, consistent with the resilience discussed for FSM-driven dialogue management. Figure 11 shows the state reached after the final refinement.

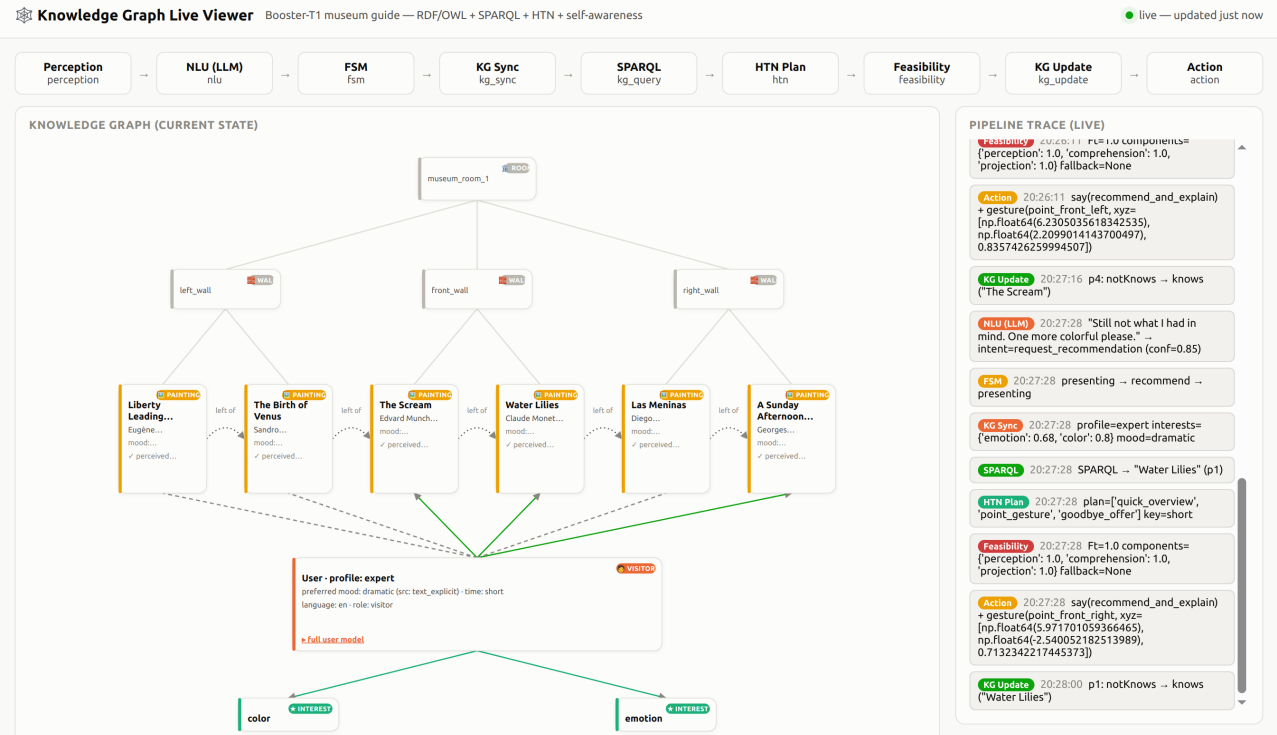


Figure 11: **Live Viewer for Scenario C — time-constrained expert, after three refinements.**

6.1.4 Scenario D: Role-Gated Operational Commands and Cross-Session Constraint Enforcement (Staff Authorization)

Visitor Persona: Giulio Bruno, a museum staff member; followed by a second, independent visitor session (played, for demonstration purposes, by the same person in front of the camera, but treated by the system as an entirely new, unrelated visitor).

The robot greets Giulio with its standard opening line. He replies: *"Hi, I'm from the museum staff."* Since no passphrase is present in this utterance, the robot asks him to verify his status: *"Welcome! To verify your staff status, please provide your security passphrase."* Giulio answers: *"The code is 1234,"* which the deterministic

passphrase check performs *before* any LLM call, flipping `role` from `visitor` to `staff`. This is the first scenario in which `role` (not `profile`) is the field being set: the two are orthogonal in the visitor model, and a staff member can in principle carry any profile value.

The NLU stage classifies this utterance as `intent=other` (the correct outcome for a bare passphrase with no accompanying instruction) and the FSM would normally react with a clarification question. Since authentication has just occurred, the robot skips that and directly acknowledges staff access instead: *"Staff access granted. Would you like me to re-scan the room to update the knowledge base, or toggle the availability of a specific painting?"*

Giulio says: *"The painting Water Lilies is not available."* The NLU returns `intent=staff_disable_painting` (`conf=0.90`); because staff operational commands are phase-independent by design (they neither require nor advance any particular phase), the FSM simply executes the command in place. This is not merely a verbal acknowledgment: the knowledge graph itself is mutated. In fact, *Water Lilies* is flagged under maintenance, rendered in the live viewer with a dashed red outline, and the robot confirms: *"Water Lilies has been marked as unavailable and will be excluded from proactive recommendations."* Giulio ends the session (*"Thank you, goodbye"* → `intent=goodbye`, `conf=0.90`, `greeting` → `goodbye` → `greeting`).

After the fixed ten-second cooldown the face-detection module enforces between conversations (`COOLDOWN_S` in `face_trigger.py`), the robot greets a new visitor. Being an entirely new session, the visitor model is reinitialized from scratch: `role` resets to `visitor` and every profile field resets to its default, regardless of what the previous visitor's session had established. This new visitor addresses the robot in Italian: *"Ciao, mi piacerebbe vedere il quadro Water Lilies,"* which the KG_SYNC stage records as `language=it`. This is the first across all three spoken scenarios where the language field takes a value other than `en`, and it does so without any explicit language selection, purely from word-level detection of the visitor's own utterance.

Naming a specific, currently unavailable painting triggers a different safety-net mechanism from the SOCIAL SUPERVISOR seen in the child scenario: a dedicated check (`_find_requested_catalog_painting`) runs immediately after intent classification but before any action is dispatched, intercepting the turn whenever a visitor names a painting that is either off display or under maintenance. The trace records this distinctly, as `greeting` → `unavailable_request("Water Lilies")`, rather than as an ordinary `action` → `next_phase` triple.

The SPARQL/KG_QUERY stage then reports *"Water Lilies" not available (maintenance)* → *propose "The Scream"*. In fact, the system searches for the closest available alternative rather than simply refusing the request, and the robot replies in Italian, explaining that the requested painting is under maintenance and offering to show *The Scream* instead.

This scenario closes the interaction walkthrough by demonstrating a key capability: operational constraints set by an authenticated staff member are not limited to a single session, but they are written into the shared knowledge graph and remain valid for all future visitors. Figure 12 shows the constraint recorded in the graph.

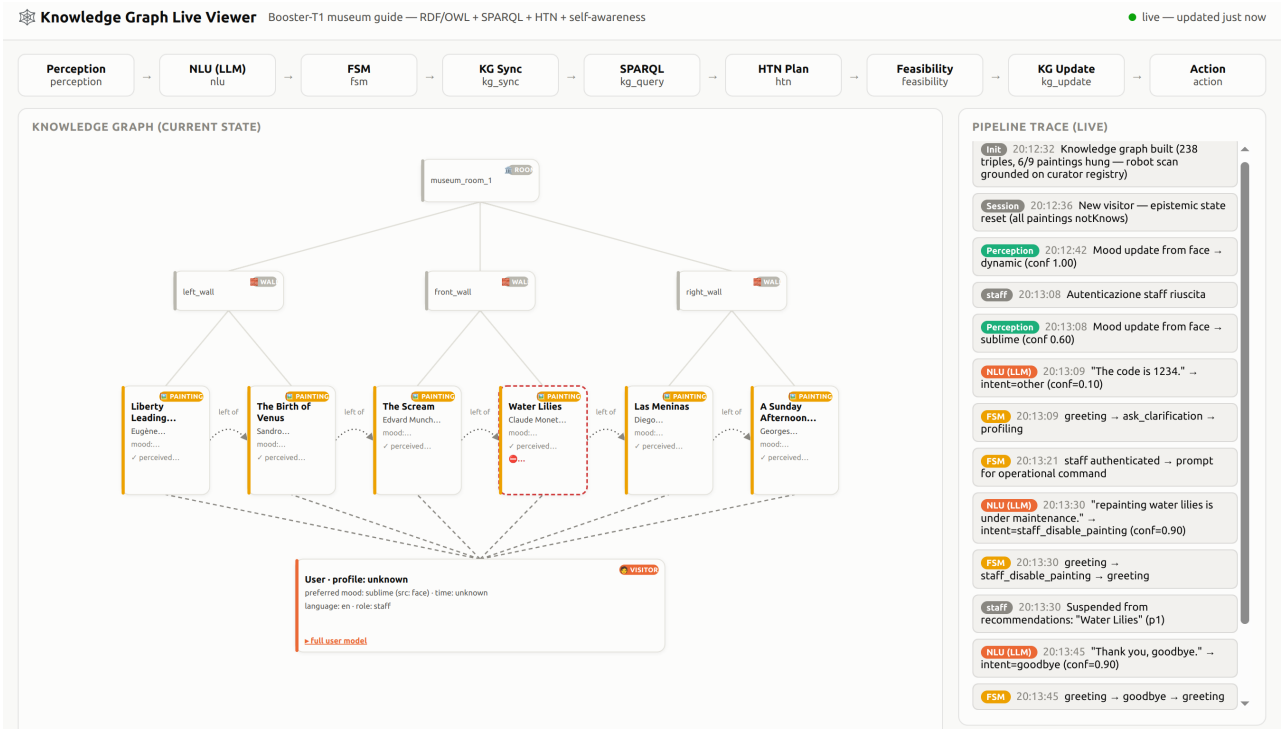


Figure 12: **Live Viewer for Scenario D — role-gated operational command.**

6.2 Quantitative Benchmark Evaluation

Complementing the qualitative walkthrough above, we ran a suite of quantitative benchmarks following the same functionality/task distinction introduced previously: functionality benchmarks isolate a single component, task benchmarks exercise the full integrated pipeline.

6.2.1 Functionality Benchmarks

Each functionality benchmark isolates one component from the rest of the pipeline (no LLM calls where not needed, no simulator where not needed):

- **Locomotion** isolates `navigate_to()`/`navigate_to_pose()` from perception and dialogue, over 4 fixed goals chosen to cover qualitatively different situations (a short hop, a long crossing, a route forced around furniture, a goal with a specific final heading). Metrics: accuracy (final position/heading error), robustness (success rate), efficiency (time, path ratio); we additionally log a person-clearance metric (minimum distance to any detected bystander).
- **Perception/VLM** isolates the single-call painting classifier (`painting_recognizer.analyze_painting`) from voting and dialogue, evaluated at each of the 6 real standing spots against the curator ground truth. Metrics: accuracy, per-spot robustness, mean call latency.

- **Face Emotion** isolates the `hsemotion-onnx` classifier on labeled static images, with no simulator involved. Metric: accuracy at two levels, the raw emotion label and the mood it maps to (a many-to-one mapping, so mood accuracy is expected to be $\geq$ raw-label accuracy).
- **Planning/IK** isolates the 2-DOF arm IK solver against real 3D target positions recovered from the last museum scan. Metrics: convergence rate, mean/max residual error, solve time.
- **Interaction/NLU** isolates `analyze_input()`'s intent classification and language detection from dialogue routing and response generation, against the fixed, hand-labeled set of 21 utterances in Table 4, chosen to cover ordinary intents alongside known edge cases. Metrics: intent accuracy, language accuracy, latency.

Utterance	**Expected Intent**	**Lang.**	**Role**
Ciao!	`greeting`	it	visitor
Hello!	`greeting`	en	visitor
Bonjour!	`greeting`	fr	visitor
Vorrei vedere un quadro colorato	`request_recommendation`	it	visitor
I'd like to see something dramatic	`request_recommendation`	en	visitor
Mi è piaciuto molto ma vorrei vedere qualcosa di più drammatico	`request_recommendation`	it	visitor
Puoi dirmi di più sulla tecnica?	`wants_more_detail`	it	visitor
Interesting, what about the artist?	`wants_more_detail`	en	visitor
Mi è piaciuto molto!	`feedback`	it	visitor
It wasn't really my style	`feedback`	en	visitor
No grazie, sono soddisfatto	`goodbye`	it	visitor
No thank you, I'm fine	`goodbye`	en	visitor
Arrivederci!	`goodbye`	it	visitor
Cos'è l'impressionismo?	`general_question`	it	visitor
What is this museum about?	`general_question`	en	visitor
sono dello staff	`other`	it	visitor
1234, riscansiona la sala	`staff_rescan`	it	staff
il quadro Water Lilies è in manutenzione	`staff_disable_painting`	it	staff
riattiva il quadro Water Lilies	`staff_enable_painting`	it	staff
Hola, quiero ver un cuadro	`request_recommendation`	es	visitor
Hallo, ich möchte ein Bild sehen	`request_recommendation`	de	visitor

Table 4: The 21 hand-labeled utterances used to evaluate the NLU functionality benchmark.

Benchmark	Key Results	Notes
Locomotion (4 goals × 2 runs)	Success: 100%, pos. error: ∼10 cm, heading error: ∼0.04 rad, path ratio: 1.07 Person clearance: mean 1.34 m, min 0.81 m (2/8 runs with a bystander present)	Clearance stays above the ∼1.2 m social-distance boundary; sample too small (bystanders were rare) to claim more than a sanity check.
Perception / VLM (6 angles × 5 runs)	100% classification accuracy, avg. latency 796 ms	Closed catalog of well-known paintings (an easy case for a modern VLM).
Face Emotion (9 images)	55.6% raw label accuracy, 77.8% mapped mood	Explainable failures (missed face crop, surprise/neutral confusion).
Planning / IK (6 targets × 10 runs)	Mean residual 2.19 cm (max 3.58 cm), 100% convergence, ∼4.7 ms	2-DOF solver, efficient enough for real-time pointing.
Interaction / NLU (21 cases × 2 runs)	100% intent & language accuracy, latency 3.4/6.1 s	Small, cleanly separated test set (Table 4).

Table 5: Functionality benchmark results (isolated components).

Locomotion, perception, and planning are all highly reliable in isolation, and the 100% scores for perception and NLU should be read with the same caution we apply elsewhere in this report: they mainly reflect a small, well-separated test catalog rather than proof of general robustness. The face-emotion result (77.8%) is, if anything, the more informative number, since it comes with concrete, explainable failure cases rather than a flat ceiling. The new person-clearance metric points the same way: encounters with a bystander were infrequent, so we report it as a preliminary observation, not a validated safety guarantee.

6.2.2 Task Benchmarks

The end-to-end scanning task (3 runs) reproduces the procedure shown in Scenario A: the robot visits every standing point, recognizes each painting, and builds the semantic map and knowledge graph incrementally. Locomotion during this integrated run reached all 6 standing points with a 100% success rate across the 3 runs, a mean position error of 12.4 cm and a mean heading error of 0.05 rad (both close to, but slightly higher than, the isolated locomotion benchmark, consistent with the longer and more turn-heavy route the full room traversal requires). The path ratio was correspondingly a little higher at 1.11 (vs. 1.07 in isolation).

Repeating the person-clearance measurement during this integrated run (rather than the isolated locomotion benchmark) gave a mean of 1.21 m and a minimum of 0.74 m over 3 recorded encounters across the 3 runs. Perception and knowledge-graph construction reached a perfect match against ground truth ($\delta = 0.0$, $M = 0.0$, $P = 0.0$) in all 3 runs, consistent with the isolated perception numbers reported above.

The conversational side was evaluated by repeating the 8 standardized visitor scripts

from `benchmark_task.py` five times each against the real pipeline, so that LLM variability shows up in the results instead of being averaged away. Table 6 reports overall and per-sub-goal success rates.

Scenario	Overall	Sub-Goal	Rate
`novice_color`	100%	`recommendation_shown` / `interest_captured`	100% / 100%
`expert_dramatic`	80%	`two_recommendations_shown` / `mood_captured`	80% / 100%
`staff_ops`	100%	`staff_authenticated` / `operational_command_executed`	100% / 100%
`social_safety_refusal`	80%	`unsafe_request_refused` / `mild_rudeness_not_overblocked` / `session_recovers`	100% / 80% / 100%
`compound_intent`	60%	`recommendation_within_budget` / `no_clarification_detour`	60% / 60%
`contradictory_mood`	60%	`mood_reversal_tracked` / `recommendation_matches_reversed_mood`	60% / 60%
`rapid_rejections`	60%	`resilient_to_rejections` / `fast_recovery`	60% / 60%
`multimodal_mood_conflict`	100%	`mood_from_face_captured` / `mood_source_correct`	100% / 100%

Table 6: Task benchmark: 8 scripted visitor scenarios, 5 repeats each.

As an example, the `staff_ops` script (the automated counterpart of Scenario D) is simply: *"1234,"* *"il quadro Water Lilies è in manutenzione,"* *"grazie, ho finito,"* fed as text with no camera/voice input.

The two perfect scenarios (`novice_color`, `multimodal_mood_conflict`) both resolve a single, unambiguous request per turn with no state to reconcile across turns.

Every scenario below 100% fails on a turn that requires carrying information forward: `social_safety_refusal`'s only failures are over-cautious refusals of mere rudeness, not missed genuine threats; `compound_intent` and `contradictory_mood` (the weakest, at 60%) both fail specifically at tracking a request or a reversed preference across turns; `rapid_rejections` fails when the system falls back into a clarification question instead of reacting directly to "no, another one".

In every case, the failure is tied to one specific, repeatable sub-goal rather than to random or unsafe output, which is the property we consider most important here.

6.3 Experimental User Evaluation

Real human participants were not available for this study, so this evaluation uses an LLM role-playing a synthetic visitor instead. After each scripted session completes end to end against the real, unmodified pipeline (`guida_museale_core.py`), the LLM

is given the assigned persona's bio and a factual summary of what actually happened in that session, and asked to fill in the Godspeed Questionnaire as that persona would. The evaluation methodology consists in designing hypotheses, running a controlled comparison and reporting a significance test.

6.3.1 Design and Hypotheses

The study uses a **between-subject** design: each synthetic participant takes part in only one condition in a single session, never both. This avoids the fatigue/practice bias a repeated-exposure design would risk, at no extra cost here since participants are generated rather than recruited.

- **H1 (Embodied Gesture):** does the robot's physical pointing gesture change how it is perceived, compared to a purely verbal description?
  IV: `gesture_present` $\in$ {`no_gesture`, `with_gesture`}.
  H1: visitors rate the robot as more intelligent and more likeable when it points at the painting. $H1_0$: no effect on any Godspeed dimension.
- **H2 (Profile Inference):** does it matter whether the visitor's profile is inferred implicitly from context or stated explicitly?
  IV: `profile_inference_mode` $\in$ {`explicit`, `implicit`}; both conditions converge on the same correct profile, only the route differs.
  H2: visitors rate the robot as more intelligent and more likeable when its profile is inferred implicitly. $H2_0$: no effect on any Godspeed dimension.

The dependent variables are the five Godspeed dimensions (anthropomorphism, animacy, likeability, perceived intelligence, perceived safety; 1–5 scale), with *likeability* and *perceived intelligence* as the primary measures for both hypotheses. $N = 12$ synthetic participants per condition (48 sessions total, 24 for H1 and 24 for H2), drawn from a fixed rotation of four visitor personas (e.g., a child with a parent, a rushed tourist, an art-history student, an older visitor) repeated identically across conditions, so persona mix cannot itself explain a difference between them.

Each synthetic session consists of:

1. a persona is assigned from the fixed rotation;
2. a fixed, scripted visitor turn sequence is fed to `guida_museale_core.py` end to end, varying only the one thing the active hypothesis manipulates;
3. a factual summary of the session (paintings shown, whether the gesture executed, turns taken, final profile) is compiled;
4. the LLM fills in the Godspeed Questionnaire in character, given the persona and that summary.

6.3.2 Results

H1 – Embodied Gesture. Figure 13 compares the two conditions across all five Godspeed dimensions. $H1_0$ is rejected for both primary dimensions: likeability rises

from 3.00 to 3.73 and perceived intelligence from 1.69 to 2.77 when the robot points at the painting. This supports our core design rationale: the gesture is not decorative, it makes the robot's spatial grounding visible, giving the visitor concrete evidence that it knows exactly where the painting is. Anthropomorphism and animacy stay essentially flat across conditions (expected, since a single rigid pointing motion is not the kind of cue that typically drives those two dimensions) and perceived safety remains high and stable in both conditions.

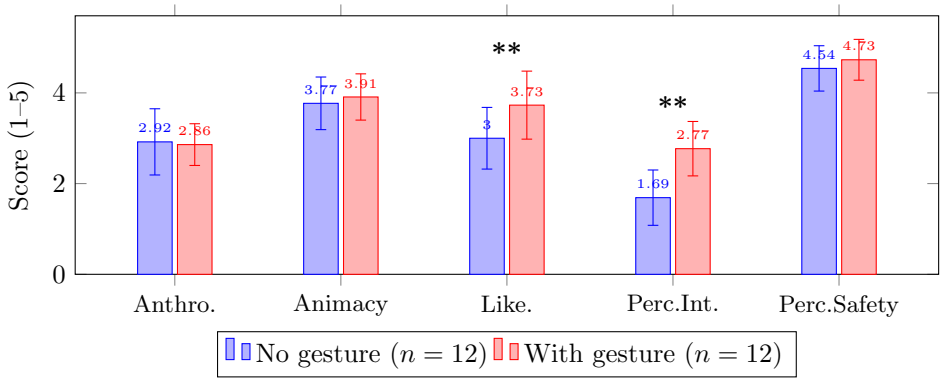


Figure 13: H1: Godspeed scores with vs. without the pointing gesture (mean ± SD). ** marks the two dimensions where $H1_0$ is rejected.

H2 – Profile Inference. Figure 14 compares explicit vs. implicit profile inference. $H2_0$ cannot be rejected for any dimension ($t \leq 1.77$ in every case). The one dimension worth commenting on is likeability, which shows a non-significant trend ($p = 0.089$) in the *opposite* direction of our hypothesis, with implicit inference (2.47) rated slightly below explicit (2.93). A plausible explanation is transparency: although the implicit inference is technically correct, the visitor never hears the robot confirm what it deduced, so they have no direct evidence of having been understood correctly. A system that acts correctly on a hidden inference may be trusted less than one that states its reasoning out loud, a concrete direction for future work (having the robot verbally confirm implicit deductions).

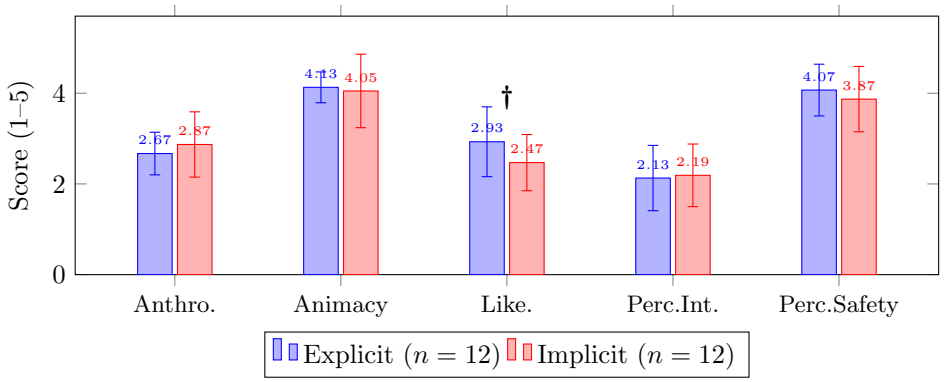


Figure 14: H2: Godspeed scores, explicit vs. implicit profile inference (mean ± SD). † marks a non-significant trend ($p = 0.089$); no dimension reaches significance.

Overall, the synthetic study confirms H1 with a clear, interpretable effect, while H2 fails to reject its null hypothesis but surfaces a counter-trend worth investigating further rather than a flat non-result.

6.3.3 Comparison with Related Work

The foundational design decision of this project (equipping the robot with an explicit pointing gesture rather than relying solely on speech) was initially justified in the Related Work by citing the embodiment lecture's premise: physical, legible action is more impactful than mere human-like appearance [5, 13]. The H1 evaluation represents the first instance in this project where this assumption is empirically tested rather than merely asserted. The results show that both primary Godspeed dimensions shift in the predicted direction when the gesture is present, providing within-project empirical support for a principle that had previously only been borrowed from literature.

The outcome for H2, however, presents a different scenario. The commonsense-reasoning lecture [18] highlights the ability to infer under-specified visitor needs as a valuable and under-explored competency, which served as the theoretical foundation for our implicit-profiling feature. Yet, H2 does not empirically confirm this advantage: implicit inference fails to outperform an explicit declaration of the same profile, and the only dimension showing a noticeable trend (likeability) actually moves against our hypothesis. While this does not invalidate the theoretical gap identified by [18], it strongly suggests that closing this gap *silently* is not automatically synonymous with closing it *usefully*.

7 Conclusions

This work presented a simulated Booster-T1 museum guide robot that combines vision-language perception, a knowledge-graph-based recommender, hierarchical task planning, and an LLM-driven dialogue manager into one integrated pipeline. We demonstrated it both qualitatively (a live, embodied video walkthrough of four representative scenarios) and quantitatively (functionality benchmarks, task benchmarks, and a synthetic Godspeed-based user evaluation).

Strengths and Limitations. The system's main strengths are its modular, independently-benchmarkable architecture; a knowledge graph that makes the visitor model and the museum's operational state explicit and queryable rather than implicit in code; a safety layer (the Social Supervisor) that runs independently of, and before, ordinary intent classification; automatic multilingual handling with no explicit language selection step; and a persistent, independently-formed robot personality that survives across sessions.

The most significant limitation is how the robot perceives people who are moving around it. Since the simulated Booster-T1 carries only an RGB-D camera and no lidar or dedicated person tracker, our perception of static and dynamic obstacles is not driven by a real sensor at all. It reads the ground-truth position of the robot, obstacles, and pedestrians directly from the simulation, and only afterwards passes it through a plausible sensor model (limited range, line-of-sight occlusion, short-term memory) to approximate what a real sensor could have measured.

This makes the demonstrated obstacle-avoidance behavior a reasonable simulation, but not something that would work on a real robot: a real deployment would need an actual sensor for detecting people (like a real-time person detector on the camera feed, or an added lidar) before any of this locomotion logic could run against real data. A secondary limitation, discussed already in Section 6's analysis, is the dialogue manager's weaker performance on scenarios that require carrying information across multiple turns rather than understanding one utterance in isolation.

Technical Difficulties. The largest source of technical difficulty in the whole project was locomotion: getting the robot to walk reliably, know where it is, and plan a path around obstacles, all made harder by the sensor limitation described above (with the initial problem of not being able to make the robot understand its position in space). Then, due to the slowness of walking, we were forced to deliberately keep the environment small (a single room) rather than a larger, multi-room museum, both to keep session and benchmark runtimes practical and because a bigger space would only have made the slow locomotion more noticeable.

The second major difficulty was external and outside of our control: the Groq-hosted LLM and VLM models we relied on were free when the project started, but were deprecated or moved behind a paywall as the project progressed, forcing us to repeatedly switch to different models mid-development. Even after we found a working free alternative, the daily quota of free tokens shared was so small that it significantly limited the amount of testing and benchmarking we could actually run daily.

Future Work. Beyond replacing the synthetic Godspeed study with a real human-participant evaluation, having the robot verbally confirm implicit inferences, and improving multi-turn dialogue state tracking, the clearest priority is making the navigation stack viable outside simulation: integrating a real sensor for detecting moving people (rather than reading their position from the simulator), and improving walking speed and path planning enough to support a larger, more realistic environment than the single room used here.

Ethical Considerations. The robot silently infers personal attributes from how a visitor speaks and looks, without ever telling them it is doing this or asking permis-

sion first. A real visitor might reasonably want to know that a camera is reading their facial expression, or feel uncomfortable being silently placed into a category like "child" or "tourist" without being asked. On the positive side, our system does not remember individual visitors between sessions, which avoids the bigger privacy issue of a robot keeping a file on people across repeat visits. Finally, the staff access control is a single fixed passphrase that anyone who happens to know it could type in. This is an acceptable simplification for a course project, but it is not how access to real museum operations should be protected in an actual deployment, which would need individual staff credentials rather than one shared code.

Generative AI Use Declaration

Generative AI was used related to the development of this project.

References

[1] Wolfram Burgard, Armin B. Cremers, Dieter Fox, Dirk Hähnel, Gerhard Lakemeyer, Dirk Schulz, Walter Steiner, and Sebastian Thrun. Experiences with an interactive museum tour-guide robot. *Artificial Intelligence*, 114(1–2):3–55, 1999.

[2] Michael A. Goodrich and Alan C. Schultz. Human-robot interaction: A survey. *Foundations and Trends in Human-Computer Interaction*, 1(3):203–275, 2007.

[3] Qiao Gu, Ali Kuwajerwala, Sacha Morin, Krishna Murthy Jatavallabhula, Bipasha Sen, Aditya Agarwal, Corban Rivera, William Paul, Kirsty Ellis, Rama Chellappa, Chuang Gan, Celso Miguel de Melo, Joshua B. Tenenbaum, Antonio Torralba, Florian Shkurti, and Liam Paull. ConceptGraphs: Open-vocabulary 3D scene graphs for perception and planning. In *IEEE International Conference on Robotics and Automation (ICRA)*, 2024. arXiv:2309.16650.

[4] Edward T. Hall. *The Hidden Dimension*. Doubleday, New York, 1966.

[5] Luca Iocchi. Human-robot interaction — lecture 2: Embodiment. Lecture slides, Elective in Artificial Intelligence, 2026. Academic year 2025/2026.

[6] Luca Iocchi. Human-robot interaction — lecture 3: Social signal processing. Lecture slides, Elective in Artificial Intelligence, 2026. Academic year 2025/2026.

[7] Luca Iocchi. Human-robot interaction — lecture 4: Motion control for HRI. Lecture slides, Elective in Artificial Intelligence, 2026. Academic year 2025/2026.

[8] Luca Iocchi. Human-robot interaction — lecture 5: Perception for HRI. Lecture slides, Elective in Artificial Intelligence, 2026. Academic year 2025/2026.

[9] Luca Iocchi. Human-robot interaction — lecture 6: Knowledge representation and planning in HRI. Lecture slides, Elective in Artificial Intelligence, 2026. Academic year 2025/2026.

[10] Luca Iocchi. Human-robot interaction — lecture 7: Evaluation. Lecture slides, Elective in Artificial Intelligence, 2026. Academic year 2025/2026.

[11] Luca Iocchi, Alessio Saladino, Vincenzo Agostiniano, Pieter Saveyn, Michele Brienza, Vincenzo Suriani, Domenico Bloisi, and Alessia Tancredi. Cognitive robot architectures through interaction of LLM-based components. Lecture slides, Elective in Artificial Intelligence, 2026. Academic year 2025/2026.

[12] Thibault Kruse, Amit Kumar Pandey, Rachid Alami, and Alexandra Kirsch. Human-aware robot navigation: A survey. *Robotics and Autonomous Systems*, 61(12):1726–1743, 2013.

[13] Jamy Li. The benefit of being physically present: A survey of experimental works comparing copresent robots, telepresent robots and virtual agents. *International Journal of Human-Computer Studies*, 77:23–37, 2015.

[14] Ali Mollahosseini, Behzad Hasani, and Mohammad H. Mahoor. AffectNet: A database for facial expression, valence, and arousal computing in the wild. *IEEE Transactions on Affective Computing*, 10(1):18–31, 2019.

[15] Jorge Rios-Martinez, Anne Spalanzani, and Christian Laugier. From proxemics theory to socially-aware navigation: A survey. *International Journal of Social Robotics*, 7(2):137–153, 2015.

[16] Allison Sauppé and Bilge Mutlu. Robot deictics: How gesture and context shape referential communication. In *ACM/IEEE International Conference on Human-Robot Interaction (HRI)*, pages 342–349, 2014.

[17] Vincenzo Suriani. HRAI — affordances and common sense. Lecture slides, Human-Robot AI Interaction, 2026. Academic year 2025/2026.

[18] Vincenzo Suriani. HRAI — commonsense. Lecture slides, Human-Robot AI Interaction, 2026. Academic year 2025/2026.

[19] Vincenzo Suriani. HRAI — towards scene graph representations. Lecture slides, Human-Robot AI Interaction, 2026. Academic year 2025/2026.

[20] Vincenzo Suriani. HRAI 2026 — geometrical and semantical representations. Lecture slides, Human-Robot AI Interaction, 2026. Academic year 2025/2026. Defines the semantic-map formalism $SM = \langle R, M, P \rangle$.

[21] Vincenzo Suriani. HRAI 2026 — intro and benchmarking. Lecture slides, Human-Robot AI Interaction, 2026. Academic year 2025/2026.

[22] Sebastian Thrun, Maren Bennewitz, Wolfram Burgard, Armin B. Cremers, Frank Dellaert, Dieter Fox, Dirk Hähnel, Charles Rosenberg, Nicholas Roy, Jamieson Schulte, and Dirk Schulz. MINERVA: A second-generation museum tour-guide robot. In *IEEE International Conference on Robotics and Automation (ICRA)*, 1999.